

Foundation (Jalapeno)

- Q1.** Factor out the greatest common factor from $12x + 16$
- (A) $4(3x + 4)$
(B) $2(6x + 8)$
(C) $8(1.5x + 2)$
(D) $12(x + \frac{4}{3})$
(E) $12x(1.67x + 1)$
- Q2.** Simplify the expression $9x^2 + 12x$ by factoring
- (A) $3x(3x + 4)$
(B) $3x(3x - 4)$
(C) $3x^2(3 + 4)$
(D) $6x(1.5x + 2)$
(E) $12x(0.75x + 1)$
- Q3.** A space probe travels $15m$ in $3s$. What is the common factor in its distance and time, and how does it relate to its speed?
- (A) The common factor is 3, which is the probe's speed in m/s
(B) The common factor is 5, which is the probe's acceleration in m/s^2
(C) The common factor is 3, which is the probe's acceleration in m/s^2
(D) The common factor is 5, which is the probe's speed in m/s
(E) There is no common factor
- Q4.** Factor the expression $20x^2 + 12x$
- (A) $2x(10x + 6)$
(B) $4x(5x + 3)$
(C) $4(5x^2 + 3x)$
(D) $2(10x^2 + 6x)$
- Q5.** A lab experiment involves mixing $8g$ of chemical A with $12g$ of chemical B. What is the greatest common factor of the masses?
- (A) $1g$
(B) $2g$
(C) $4g$
(D) $8g$
(E) $12g$
- Q6.** The planet Mars has a diameter of $6794km$. A scale model of Mars has a diameter of $27cm$. What is the common factor in their diameters?
- (A) 1
(B) 3
(C) 9
(D) 27
(E) 81
- Q7.** Factor out the greatest common factor from $18x^2 + 24x + 6$
- (A) $6(3x^2 + 4x + 1)$
(B) $6(3x^2 - 4x - 1)$
(C) $3(6x^2 + 8x + 2)$
(D) $2(9x^2 + 12x + 3)$
(E) $12(1.5x^2 + 2x + 0.5)$
- Q8.** A physics equation involves the expression $24x + 36$. Factor out the greatest common factor
- (A) $12(2x + 3)$
(B) $12(2x - 3)$
(C) $6(4x + 6)$
(D) $4(6x + 9)$
(E) $3(8x + 12)$

Open Ended Questions (FRQ)

- Q9.** A space station has 15 modules, each with a capacity of 8 astronauts. Factor out the greatest common factor from the total capacity and the number of modules, and explain the significance of the result in 5 lines or less
- Q10.** The orbit of a planet can be modeled using the equation $12x^2 + 16x$. Factor out the greatest common factor and simplify the expression. Show all work and explain the result in 5 lines or less

Chef's Tips

- Tip 1: Encourage students to identify the greatest common factor by finding the largest number that divides both terms
- Tip 2: Remind students to simplify expressions by factoring out the greatest common factor
- Tip 3: Emphasize the importance of factoring in real-world applications, such as physics and engineering